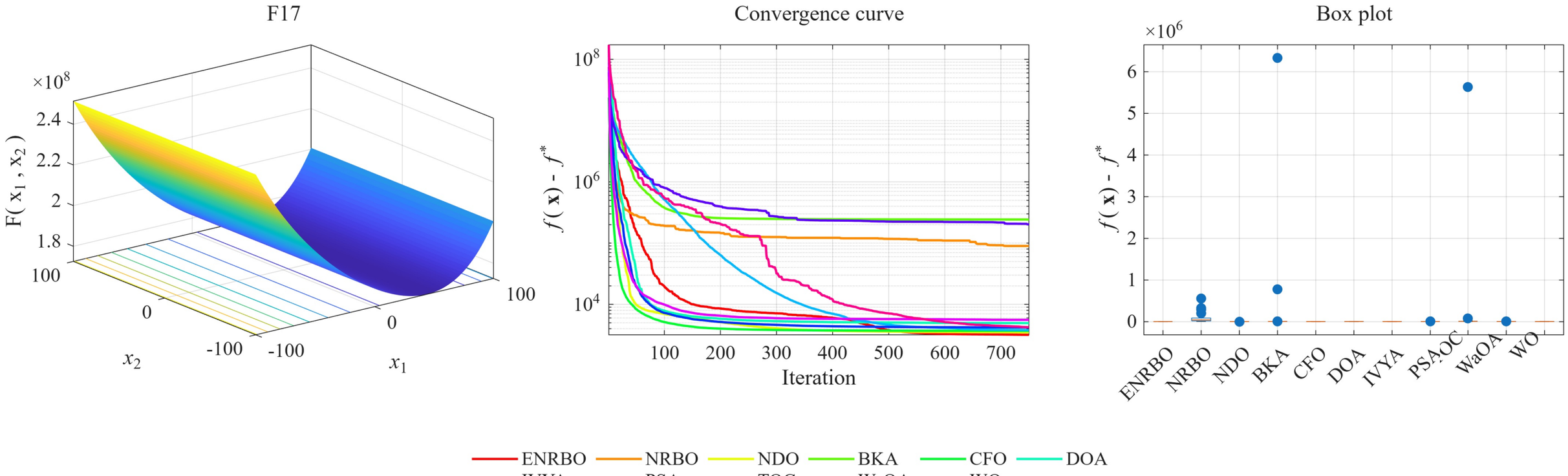

F17
×10^8
2.4
2.2
2
1.8
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^8
10^6
10^4
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^6
6
5
4
3
2
1
0
f(x) - f*
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
OC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO